PII Redaction Engine: Evaluation Strategy & Metrics

1. Primary Objective

The goal of this redaction system is to maximize recall by identifying and sanitizing all personally identifiable information (PII) while maintaining precision so that standard legal, financial, and regulatory terms remain unredacted.

2. Evaluation Strategy

The evaluation focused on handling edge cases and structural anomalies commonly found in Microsoft Word (.docx) XML documents:

Visual PII Sanitization: Embedded images (such as scanned PAN or Aadhaar cards) can bypass text-based extraction. The system inspects package image streams and replaces binary blobs with a 1x1 transparent image, preventing visual data leaks.

Formatting & Hidden Character Handling: Word XML files often contain soft returns, tabs, and mid-sentence tags that break simple word-boundary matching. The pipeline uses whitespace-flexible regex and gazetteer matching to detect entities fragmented across formatting tags.

False-Positive Control: To avoid over-redaction by the SpaCy NER model, a domain protected vocabulary whitelist is checked before replacement. Key terms such as "Offer for Sale", "Qualified Institutional Buyers", and "SEBI" are preserved.

3. Core Metrics

Recall: High recall is prioritized to ensure complete sanitization. Specific entities (promoters, banks, identifiers) are targeted via deterministic gazetteer and regex matching before applying NER.

Precision: Precision is maintained by combining context rules and protected term whitelists, differentiating sensitive organizations from standard regulatory bodies.

F1-Score: The F1-score reflects the balance between aggressive identifier detection and protected vocabulary filtering.